

# Syllabus for

Academic year 2020/2021 &gt;

## TMA672 - Linear algebra and numerical analysis

*Linjär algebra och numerisk analys*

Syllabus adopted 2020-02-12 by Head of Programme (or corresponding)

Owner: [TKTFY](#)

**7,5 Credits**

**Grading:** TH - Pass with distinction (5), Pass with credit (4), Pass (3), Fail

**Education cycle:** First-cycle

**Major subject:** Mathematics, Engineering Physics

**Department:** 11 - MATHEMATICAL SCIENCES

**Teaching language:** Swedish

**Application code:** 57117

**Open for exchange students:** No

Maximum participants: 190

Only students with the course round in the programme plan

### Credit distribution

| Module                             | Sp1 | Sp2 | Sp3 | Sp4   | Summer course  | No Sp | Examination dates  |
|------------------------------------|-----|-----|-----|-------|-------------------------------------------------------------------------------------------------|-------|-------------------------------------------------------------------------------------------------------|
| 0119 Laboratory 1,5 c Grading: UG  |     |     |     | 1,5 c |                                                                                                 |       |                                                                                                       |
| 0219 Examination 6,0 c Grading: TH |     |     |     | 6,0 c |                                                                                                 |       | Mon 31/05-2021 pm J, Sat 10/10-2020 am J, Thu 26/08-2021 am J                                         |

### In programs

[TKAUT AUTOMATION AND MECHATRONICS ENGINEERING, Year 3 \(elective\)](#)

[TKTEM ENGINEERING MATHEMATICS, Year 1 \(compulsory\)](#)

[TKTFY ENGINEERING PHYSICS, Year 1 \(compulsory\)](#)

### Examiner:

[Thomas Bäckdahl](#)



[Go to course homepage](#)

### Eligibility

General entry requirements for bachelor's level (first cycle)

Applicants enrolled in a programme at Chalmers where the course is included in the study programme are exempted from fulfilling the requirements above.

### Specific entry requirements

The same as for the programme that owns the course.

Applicants enrolled in a programme at Chalmers where the course is included in the study programme are exempted from fulfilling the requirements above.

### Course specific prerequisites

A first course in Linear algebra and geometry

### Aim

To present an introduction to linear spaces concepts such as linear dependence/independence, basis, dimension, orthogonality and spectral theory.

To give the basics in numerical analysis and computational mathematics.

### Learning outcomes (after completion of the course the student should be able to)

- use linear algebra concepts for solving problems in engineering and science.
- use mathematical models for numerical solution of real world problems.
- critically analyze and give advice regarding different methods and algorithms with respect to efficiency and reliability.

### Content

- Vector spaces: Basis, dimension, sub-spaces.
- Inner product, orthogonality.
- Linear mappings.
- Eigenvalues and eigenvectors, spectral theory, quadratic forms.
- Applications in analysis; systems of ordinary differential equations.
- Numerical techniques for solution of linear systems, least squares problems and eigenvalue problems.
- QR-factorization and SVD.
- Numerical equation solving and optimization, search methods.
- Interpolation, and its use for numerical integration and differentiation, splines.
- Numerical solution of differential equations, explicit and implicit methods, stability.

**Organisation**

Lectures, lessons, and computer exercises.

**Literature**

- Kjell Holmåker och Ivar Gustafsson, Linjär Algebra: fortsättningskurs, 1st ed., Liber
- Ivar Gustafsson och Kjell Holmåker, Numerisk Analys, 1st ed., Liber

**Examination including compulsory elements**

Written examination with problems of theoretical and applied nature.

Computer exercises. Homework assignments giving credit points for the examination may exist.